Practical No. 2 & 3

Demonstrate to use of Version Control System (Git offline and connect to github account).
Create multiple users and usage with team leader role and coder roles.
Demonstrate code pull and push, fork (branching) amongst the users
Compare it with hg. (on answer sheet)

Demonstrate to use of Version Control System (Git offline and connect to github account).
Create multiple users and usage with team leader role and coder roles.
Demonstrate code pull and push, fork (branching) amongst the users, code merge, code diff amongst the users.
Compare git with gitea. (on answer sheet)

Procedure

Step 1: Create Repository on GitHub

Create a repository named:

test

Copy the repository URL.

Step 2: Clone Repository

```
git clone https://github.com/username/test.git

cd test
```

Step 3: Team Leader Work

```
echo "Project created by Team Leader" > leader.txt

git add .
git commit -m "Initial commit by Team Leader"
```

git push origin main

Step 4: Create Branch for Coder1

git checkout -b coder1

echo "Feature developed by coder1" > coder1.txt

git add .
git commit -m "Coder1 work"

git push origin coder1

Step 5: Create Branch for Coder2

git checkout main

git checkout -b coder2

echo "Feature developed by coder2" > coder2.txt

git add .
git commit -m "Coder2 work"

git push origin coder2

Step 6: Demonstrate Pull

git checkout main

git pull origin main

Step 7: Display Branches

git branch

Step 8: Demonstrate Push

echo "Final update by Team Leader" >> leader.txt

git add .

```
git commit -m "Final update"

git push origin main
```

Git	Mercurial (HG)
Developed by Linus Torvalds	Developed by Matt Mackall
Distributed Version Control System (DVCS)	Distributed Version Control System (DVCS)
Most widely used VCS worldwide	Less widely used
Used by GitHub	Not natively used by GitHub
Faster for large projects	Good performance but less optimized for huge projects
Large open-source community	Smaller community
Powerful branching and merging	Simpler branching and merging
More commands and flexibility	Easier to learn
Supported by many DevOps tools	Limited tool integration compared to Git
Industry standard for software development	Used in fewer organizations

Practical No.3:

Step 6: Demonstrate Diff

Compare main and coder1:

git diff main coder1

Compare coder1 and coder2:

git diff coder1 coder2

Step 7: Merge coder1

git checkout main

git merge coder1 -m "Merged coder1 into main"

Step 8: Merge coder2

git merge coder2 -m "Merged coder2 into main"

Step 9: Push Merged Result

git push origin main

Step 10: Show Commit History

git log --oneline --graph --all

Step 11: Verify Files

dir

Files present:

leader.txt

coder1.txt

Coder2.txt

If you're inside Vim

1. Press:

Esc

2. Type:

:wq

3. Press:

Enter

This means:

- w = write/save
- q = quit

Git will complete the merge.

Git	Gitea
Version Control System (VCS)	Self-hosted Git repository management platform
Used to track source code changes	Used to host and manage Git repositories
Command-line based tool	Web-based interface
Developed by Linus Torvalds	Developed by the Gitea Project
Performs commit, merge, branch, diff operations	Provides UI for repositories, issues, pull requests, and users

Can work completely offline	Usually runs as a server
Core version control technology	Uses Git internally
No repository hosting features	Provides repository hosting features
No user management interface	Supports user and team management
No built-in issue tracking	Includes issue tracking and project management features